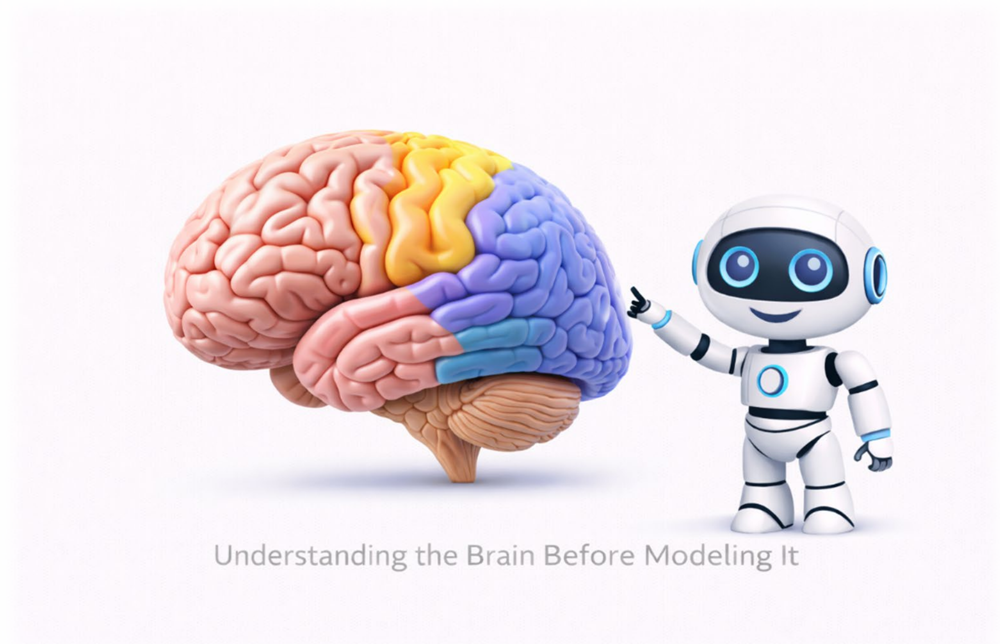
MODULE 03

BRAIN ANATOMY

&

COMPUTATIONAL NEUROSCIENCE

Understanding the Brain Before Modeling It



ITAI 4374: Neuroscience as a Model for AI

Houston City College • BAT in AI & Robotics

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Student Booklet — Restructured Edition

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PART 1

THE BRAIN

What You Need to Know First (No Math Required)

🎯 WHAT YOU'LL LEARN IN PART 1

Before we can model the brain mathematically, we need to understand what we're modeling!

In Part 1, you will learn:

- What the major parts of the brain are and what they do
- How the brain is organized (hint: like software with layers)
- What a neuron looks like and its basic parts
- How neurons send signals to each other (the basics, no equations yet)

This is the foundation. Part 2 will build mathematical models on top of this knowledge.

💡 WHY START WITH ANATOMY?

As AI students, you already know that to build a good model, you need to understand what you're modeling.

You wouldn't build a flight simulator without understanding how planes work.

You wouldn't build a weather model without understanding atmospheric physics.

You can't build brain-inspired AI without understanding the brain.

Don't worry — you don't need to become a biologist. You just need enough understanding to know what the mathematical models are trying to capture.

SECTION 1

Introduction and Course Context

1.1 Where We Are in the Course

Welcome to Module 03 of ITAI 4374: Neuroscience as a Model for AI. This module is divided into two clear parts to help you build understanding step by step.

🎯 **NO PRIOR KNOWLEDGE NEEDED! You don't need any background in biology, neuroscience, or advanced math. We'll explain everything from scratch using everyday examples. If you can understand how a smartphone battery works, you can understand neurons!**

In Module 01, you learned WHY we study neuroscience as AI engineers — the brain's remarkable efficiency (20 watts vs. megawatts for AI), its ability to learn from few examples, and its common sense reasoning that AI still lacks.

In Module 02, you learned WHAT the brain computes — predictions, world models, minimizing surprise. You studied theories like Thousand Brains, Predictive Coding, Free Energy, and Global Workspace.

But those theories were abstract. WHERE in the brain do these computations happen? HOW does a neuron actually process information? That's what Module 03 answers.

✓ COMPLETED	▶ YOU ARE HERE	NEXT →
Module 02	**Module 03**	**Module 04**
WHAT does the brain compute?	*WHERE and HOW?*	*Networks that LEARN*

1.2 How This Module is Organized

This booklet is divided into two parts:

PART 1: THE BRAIN (Sections 1-5) covers anatomy and biology. No math. No equations. Just understanding what the brain looks like, what the parts are called, and what they do. Think of this as learning the 'hardware specification.'

PART 2: COMPUTATIONAL NEUROSCIENCE (Sections 6-9) covers how we model the brain mathematically. This is where the equations come in — but we'll explain every equation in plain English. Think of this as learning to write 'software' that mimics the hardware.

📖 **STUDY TIP: If any equation feels confusing, STOP and draw a picture of what it describes. Draw the leaky bucket. Draw the voltage going up and down. The visual will help the math click!**

If you find Part 2 challenging, come back and review Part 1. The math will make more sense when you can visualize what it's describing.

SECTION 2

The Big Picture — Brain Overview

2.1 What Does a Brain Look Like?

Before diving into details, let's get oriented. The human brain is roughly the size of two fists held together, weighs about 1.4 kg (3 pounds), and has a wrinkled, pinkish-gray appearance.

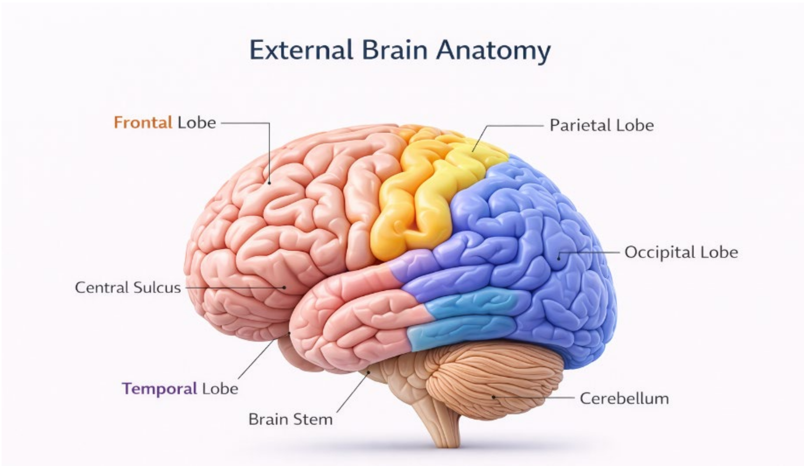


Figure 2.1: External Brain Anatomy — Lateral view showing the four lobes of the cerebral cortex.

The brain's wrinkled surface (like a walnut) isn't random — it's a way to pack more surface area into a limited space. If you flattened out the human cortex, it would be about the size of a large pizza. All those folds let it fit inside your skull.

🔢 **THE BRAIN BY THE NUMBERS**

These numbers will help you appreciate the scale of what we're studying:

- ~86 billion neurons (the cells that process information)
- ~600 trillion synapses (connections between neurons)
- ~20 watts power consumption (less than a light bulb!)

- ~100 billion glial cells (support cells, not covered in this module)

For comparison:

- GPT-4 training: estimated millions of watts
- Your laptop: 50-100 watts
- A fruit fly brain: ~100,000 neurons
- A mouse brain: ~70 million neurons

2.2 Three Major Divisions

The brain evolved in layers over millions of years. Think of it like software with legacy code at the bottom and newer features on top:

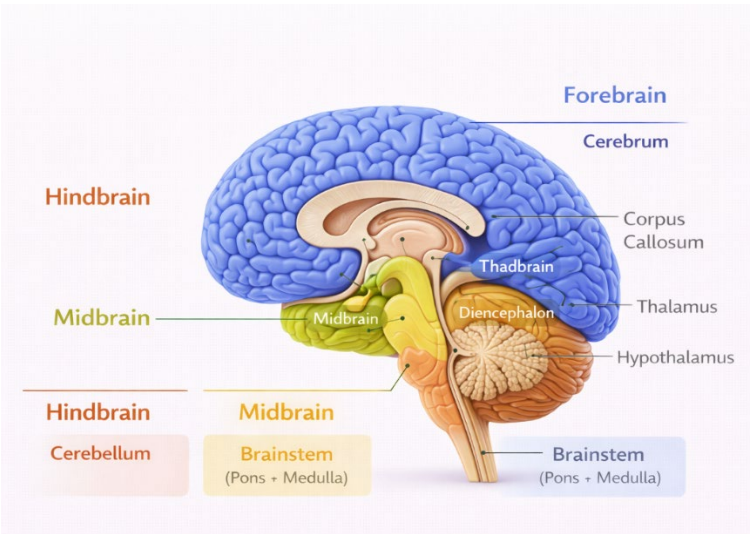


Figure 2.2: Brain Divisions (Sagittal View) — The three major divisions: hindbrain, midbrain, and forebrain.

Brain Divisions (Sagittal View)

HINDBRAIN (Oldest — Reptilian Brain)

Located at the base of the brain, connecting to the spinal cord.

What it controls: Breathing, heart rate, blood pressure, sleep/wake cycles, balance, coordination

Key structures: Medulla, Pons, Cerebellum

AI analogy: This is like your computer's BIOS — basic functions that must always run, even when you're 'asleep.'

🦑 REAL LIFE EXAMPLE: Ever noticed you keep breathing when you're asleep? Or that your heart beats without you thinking about it? That's your hindbrain working 24/7 — it's been keeping your ancestors alive for millions of years!

MIDBRAIN (Middle Layer)

A small region between the hindbrain and forebrain.

What it controls: Eye movements, auditory reflexes, dopamine production

Key structures: Superior and inferior colliculi, substantia nigra

AI analogy: Like interrupt handlers — quick reflexive responses before the 'main program' (cortex) gets involved.

FOREBRAIN (Newest — Mammalian Brain)

The large, wrinkled part that makes up most of what you see when you look at a brain.

What it controls: Thinking, planning, language, memory, perception, consciousness

Key structures: Cerebral cortex (the wrinkled outer layer), thalamus, hypothalamus, hippocampus, amygdala

AI analogy: This is where the 'high-level reasoning' happens — your transformer models, your planning algorithms, your working memory.

> **💡 THINK ABOUT IT: Why This Layered Structure?**
>
> Evolution doesn't start from scratch — it builds on what exists.
>
> When mammals evolved, they didn't replace the reptilian brain. They kept it (you still need to breathe!) and added new structures on top.
>
> This is like software development: you don't rewrite the operating system every time you add a new app. You build on top of existing, tested code.
>
> This has implications for AI: maybe we shouldn't try to build one monolithic 'intelligence' but instead layer specialized systems that work together.

SECTION 3

Brain Regions and Their Functions

3.1 The Four Lobes of the Cerebral Cortex

The cerebral cortex — the wrinkled outer layer — is divided into four lobes. Each lobe has specialized functions, though they all work together.

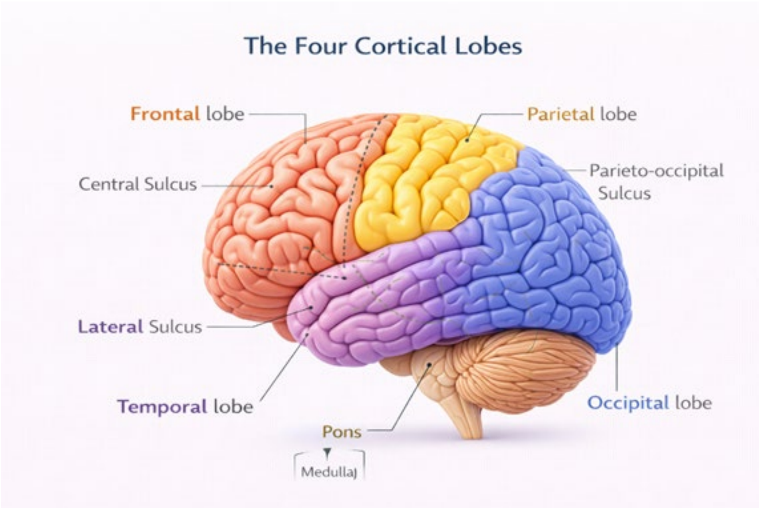


Figure 3.1: The Four Cortical Lobes — Frontal (planning), parietal (senses), temporal (hearing/memory), occipital (vision).

FRONTAL LOBE (Front of the brain)

Location: Behind your forehead, extending to the top of your head

Functions:

• Planning and decision-making (especially the **prefrontal cortex (PFC)** — the front-most part)

• Working memory (holding information in mind)

• Personality and social behavior

REAL LIFE EXAMPLE: When you resist eating the last slice of pizza because you're saving it for your roommate, that's your frontal lobe controlling impulses! When you plan out your study schedule for finals week, that's also your frontal lobe at work.

• Motor control (the back part controls movement)

• Speech production (Broca's area, left side)

🏥 CLINICAL INSIGHT: Phineas Gage

In 1848, railroad worker Phineas Gage survived an iron rod passing through his frontal lobe.

He could walk and talk, but his personality changed dramatically — he became impulsive, unreliable, and made poor decisions.

This case taught us that the frontal lobe is crucial for personality and judgment, not just movement.

PARIETAL LOBE (Top-back of the brain)

Location: Behind the frontal lobe, on top of the head

Functions:

• Processing touch, temperature, and pain

• Spatial awareness ('where am I? where are objects?')

• Coordinate transformations (converting 'what I see' to 'where to reach')

• Mathematical reasoning

• Attention

TEMPORAL LOBE (Sides of the brain)

Location: On the sides, roughly behind your ears

Functions:

• Hearing and sound processing

• Language comprehension (Wernicke's area, left side)

• Object recognition ('what is this thing?')

• Memory formation (works with hippocampus)

• Emotional processing (works with amygdala)

OCCIPITAL LOBE (Back of the brain)

Location: At the very back of the head

Functions:

• ALL visual processing

• Recognizing colors, shapes, motion

• Processing visual information before sending it forward

> 🤖 **AI CONNECTION: The Visual Processing Hierarchy**
>
> The occipital lobe processes vision in stages:
>
> - V1 (primary visual cortex): Detects edges and orientations
> - V2: Processes contours and simple shapes
> - V4: Processes color and more complex shapes
> - IT (inferotemporal cortex, actually in temporal lobe): Recognizes objects
>
> Sound familiar? This is EXACTLY what a Convolutional Neural Network (CNN) does!
>
> Early layers → edges, Later layers → shapes, Final layers → objects
>
> This is not a coincidence. CNNs were directly inspired by Hubel & Wiesel's Nobel Prize-winning discovery of how the visual cortex works.
>
> This hierarchical processing was discovered by **Hubel and Wiesel** in the 1950s-60s (Nobel Prize, 1981). Their work directly inspired Convolutional Neural Networks!

3.2 Subcortical Structures

Below the wrinkled cortex lie several crucial structures. You can't see them from the outside, but they're essential for brain function.

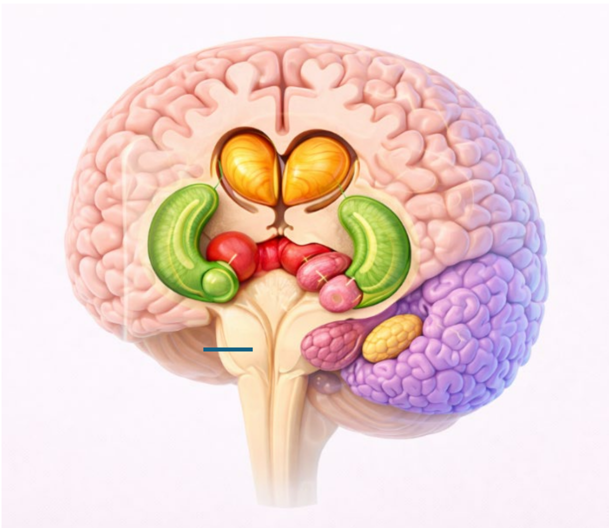


Figure 3.2: Subcortical Structures — Thalamus, hippocampus, amygdala, and basal ganglia.

Subcortical Structures — Color Key

Orange → **Thalamus:** Central relay structure near the midline ,

Red → **Hypothalamus:** Small structure inferior and medial to the thalamus

Green → **Basal Ganglia (simplified):** Striatum + Globus Pallidus, lateral to the thalamus

Purple → **Hippocampus:** Curved structure in the medial temporal lobe

Yellow → **Amygdala**

THALAMUS — The Brain's Router

Location: Center of the brain, shaped like two small eggs

Function: Almost ALL sensory information passes through here before reaching the cortex (except smell). It's not just a relay — it actively filters and gates information.

AI analogy: Like a network router or attention mechanism that decides what information gets broadcast widely.

HIPPOCAMPUS — The Memory Writer

Location: Deep in the temporal lobe, shaped like a seahorse (that's what 'hippocampus' means in Greek)

Function: Essential for forming NEW memories. Also involved in spatial navigation ('place cells' that fire at specific locations).

AI analogy: Like a write buffer that consolidates new information into long-term storage. Experience replay in deep RL was inspired by how the hippocampus replays experiences during sleep!

💥 REAL LIFE EXAMPLE: Remember how you got to school today? That memory was formed by your hippocampus! It's like your brain's 'Save' button — without it, new experiences would vanish like a Snapchat message.

> **🏥 CLINICAL INSIGHT: Patient H.M.**
>
> In 1953, Henry Molaison (known as H.M.) had both hippocampi surgically removed to treat severe epilepsy.
>
> Result: He could no longer form NEW memories. He met his doctors fresh every single day for 55 years.
>
> But he could still recall memories from BEFORE the surgery, and he could still learn new motor skills (like mirror drawing).
>
> This taught us that the hippocampus is needed for FORMING new memories, not storing old ones. And that there are different memory systems.

AMYGDALA — The Emotional Tagger

Location: Near the hippocampus, almond-shaped (that's what 'amygdala' means)

Function: Processes emotions, especially fear. Tags memories with emotional significance ('remember this — it was important/dangerous/rewarding').

AI analogy: Like a priority system that flags certain experiences for special attention and deeper encoding.

BASAL GANGLIA — The Action Selector

Location: Deep in the brain, a group of interconnected structures

Function: Selects which actions to perform, forms habits, involved in reward-based learning.

AI analogy: Like an actor-critic algorithm in reinforcement learning. The cortex proposes actions (actor), the basal ganglia evaluates them based on expected reward (critic).

> 🤖 **AI CONNECTION: Dopamine = TD Error**
>
> Schultz, Dayan & Montague (1997) made a remarkable discovery:
>
> Dopamine neurons in the basal ganglia fire exactly like the temporal difference (TD) error in reinforcement learning!
>
> - Unexpected reward → dopamine SPIKE (positive error)
> - Expected reward → no change (error = 0)
> - Missing expected reward → dopamine DIP (negative error)
>
> The brain invented reinforcement learning millions of years before computer scientists did!

SECTION 4

The Neuron — The Brain's Building Block

4.1 What is a Neuron?

Everything we've discussed so far — the lobes, the structures, the functions — is implemented by networks of neurons. A neuron is a single brain cell that can receive signals, process them, and send signals to other neurons.

Think of neurons as the 'transistors' of the brain. A single transistor doesn't do much, but billions of them connected together can run complex software. Similarly, a single neuron is simple, but 86 billion of them connected by 600 trillion synapses create human intelligence.

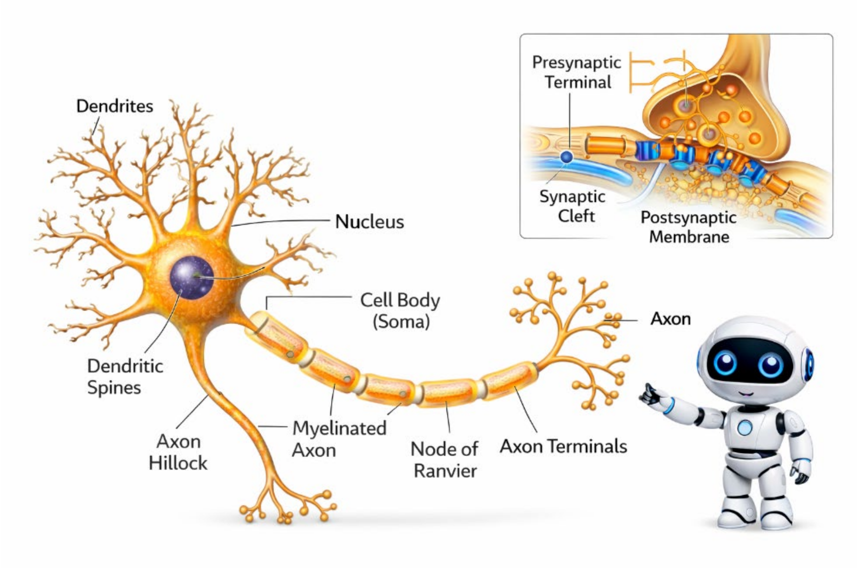


Figure 4.1: Neuron Anatomy — Dendrites (input), soma (cell body), axon (output), and synaptic terminals.

Neuron Anatomy

4.2 Parts of a Neuron

DENDRITES — The Inputs

What they look like: Tree-like branches extending from the cell body

What they do: Receive signals from other neurons. A single neuron can have thousands of dendrites receiving input from thousands of other neurons.

AI analogy: Like the input connections in an artificial neural network node.

CELL BODY (SOMA) — The Processor

What it looks like: The central 'blob' of the neuron, containing the nucleus

What it does: Integrates (adds up) all the signals coming from the dendrites. Decides whether to fire a signal or not.

AI analogy: Like the weighted sum + activation function in an ANN node.

AXON — The Output

What it looks like: A long, thin fiber extending from the cell body. Can be very long — some axons run from your spine to your toes (over a meter)!

What it does: Carries the output signal (called an action potential or 'spike') to other neurons.

AI analogy: Like the output connection from an ANN node.

SYNAPSE — The Connection Point

What it is: The junction between one neuron's axon and another neuron's dendrite

What it does: Converts the electrical signal into a chemical signal (neurotransmitters) and back to electrical. This is where learning happens — synapses can get stronger or weaker.

AI analogy: Like the 'weight' in an ANN connection. Learning = changing synaptic strengths.

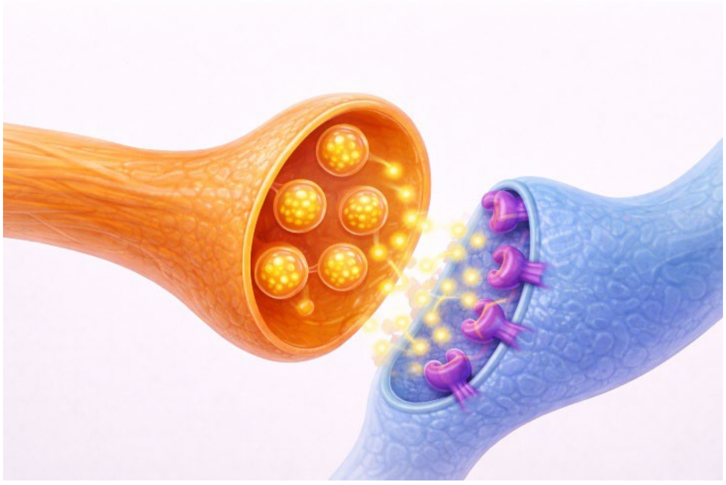
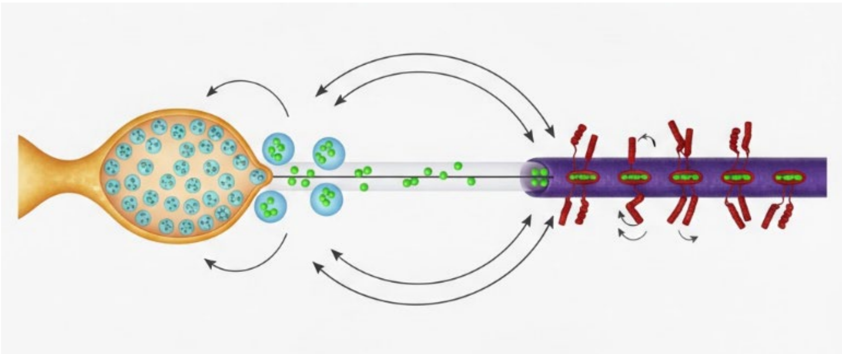


Figure 4.2: The Synapse — Neurotransmitter release across the synaptic cleft. See legend below.



Synapse Diagram Legend

Feature	Description	Role in Biological AI
Bulbous Structure (Left)	Presynaptic Terminal: The end of an axon where signals originate.	Output Layer: Represents the signal being sent from one "neuron" to the next.
Blue Spheres (Inside Bulb)	Synaptic Vesicles: Small sacs that store chemical messengers (neurotransmitters).	Data Packets: The raw information or signal values prepared for transmission.
Green Spheres (Released)	Neurotransmitters: Chemicals that transmit signals across the gap.	Signal Value: The actual numerical data being passed through the network.
Central Gap	Synaptic Cleft: The tiny physical space between the two neurons.	Transmission Step: The point in a computational graph where data moves between nodes.
Purple Structure (Right)	Postsynaptic Membrane: The receiving surface of the next neuron (dendrite).	Input Layer: The receiving end of the subsequent layer in an Artificial Neural Network.
Red/Helix Structures	Receptors: Proteins that "capture" the neurotransmitters to trigger a response.	Activation Function: The mathematical threshold (like ReLU or Sigmoid) determining if the signal continues.
Arrows	Signal Direction: Shows the flow of neurotransmitters from sender to receiver.	Information Flow: Represents the forward propagation of data in a model.

⚠️ **MISCONCEPTION ALERT: ANNs ≠ Real Neurons**

You might think: 'I already know neurons — they're like ANN nodes!'

But biological neurons are VERY different:

- ANN nodes output continuous values (0.73). Real neurons output discrete SPIKES (on/off).
- ANN nodes are instantaneous. Real neurons have TIMING — when a spike happens matters.
- ANN nodes are simple sums. Real neurons have complex DYNAMICS — they 'remember' recent inputs.
- ANNs use backpropagation. Real brains use LOCAL learning rules.

Part 2 will explore these differences in detail!

SECTION 5

How Neurons Communicate

5.1 The Action Potential — A Spike

When a neuron 'fires,' it generates an electrical pulse called an action potential (or 'spike'). This is the fundamental unit of communication in the brain.

The Basic Idea (No Math Yet)

Think of a neuron like a water balloon with a trigger:

🧪 *TRY THIS AT HOME: Get a balloon and slowly fill it with water from the tap. Notice how nothing happens for a while, then suddenly — POP! That 'nothing... nothing... BOOM!' pattern is exactly how neurons work. There's no '50% pop' — it either pops or it doesn't. This is the 'all-or-nothing' principle!*

- The neuron starts at a resting state (balloon is stable)
- Inputs from other neurons slowly 'fill' it (voltage rises)
- If enough water accumulates, the balloon pops (threshold reached → SPIKE!)

- After popping, it resets and starts over (refractory period)

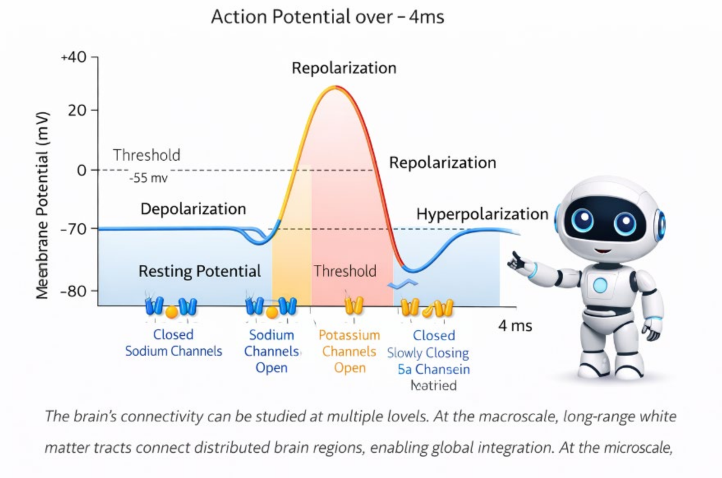


Figure 5.1: The Action Potential — Voltage changes during a neural spike.

Key Properties of Spikes

ALL-OR-NOTHING: A spike either happens or it doesn't. There are no 'half spikes' or '73% spikes.' This is very different from ANNs, where a node can output any value between 0 and 1.

FAST: A single spike lasts only 1-2 milliseconds. Neurons can fire up to 500-1000 times per second.

REFRACTORY PERIOD: After firing, a neuron briefly CAN'T fire again (absolute refractory, ~1-2 ms), then can only fire if the input is extra strong (relative refractory, ~2-4 ms). This limits maximum firing rate.

5.2 What Triggers a Spike?

A spike happens when enough inputs arrive to push the neuron's voltage above a threshold. Let's break this down:

RESTING POTENTIAL: When nothing is happening, a neuron's inside is about -70 millivolts (mV) compared to outside. Think of this as the 'default' state.

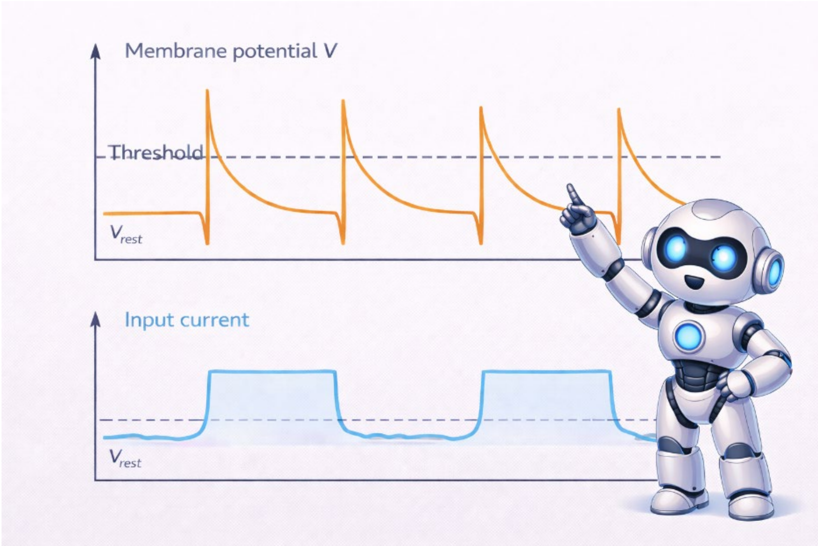
EXCITATORY INPUTS: Some synapses push the voltage UP (toward 0 and beyond). These are like 'votes' for the neuron to fire.

INHIBITORY INPUTS: Some synapses push the voltage DOWN (more negative). These are like 'votes' against firing.

THRESHOLD: If the voltage reaches about -55 mV, BOOM — a spike fires automatically. The neuron doesn't 'decide' to fire; it's automatic once threshold is reached.

☑ QUICK CHECK: If resting voltage is -70mV and threshold is -55mV, how much does the voltage need to rise to trigger a spike? Answer: 15mV (because -70 + 15 = -55). See, you just did neuroscience math!

Figure 5.2: Threshold and Reset — Spike fires when membrane potential reaches threshold.



🔑 KEY CONCEPT: Integration

A neuron doesn't just pass along signals — it INTEGRATES them.

Imagine you're deciding whether to see a movie:

- Friend A says it's great (+1 vote)
- Friend B says it's terrible (-1 vote)
- Reviews say it's good (+1 vote)
- It's expensive (-1 vote)

You add up all the votes. If the total exceeds your threshold, you go see it.

Neurons work similarly: they add up excitatory (+) and inhibitory (-) inputs, and if the total exceeds threshold, they fire.

5.3 Summary of Part 1

Before moving to Part 2, make sure you understand these key concepts:

• The brain has three major divisions (hindbrain, midbrain, forebrain) that evolved in layers

• The cerebral cortex has four lobes (frontal, parietal, temporal, occipital) with specialized functions

• Subcortical structures (thalamus, hippocampus, amygdala, basal ganglia) handle routing, memory, emotion, and action selection

• Neurons have dendrites (inputs), soma (processor), axon (output), and synapses (connections)

• Neurons communicate via action potentials (spikes) — all-or-nothing electrical pulses

• A spike fires when inputs push voltage above threshold

🎉 CONGRATULATIONS! You just completed Part 1! You now know more about the brain than 99% of people. Take a 5-minute break — your brain actually uses 20% of your body's energy, so grab a snack! Then come back ready for Part 2.

> ✓ **KNOWLEDGE CHECK: Ready for Part 2?**
>
> Can you answer these questions?
>
> 1. What are the four lobes of the cortex and what does each do?
> 2. What structure is essential for forming new memories?
> 3. What are the four main parts of a neuron?
> 4. What is the 'all-or-nothing' property of spikes?
> 5. What happens when a neuron's voltage reaches threshold?
>
> **If you're unsure about any of these, review the relevant section before continuing to Part 2.**

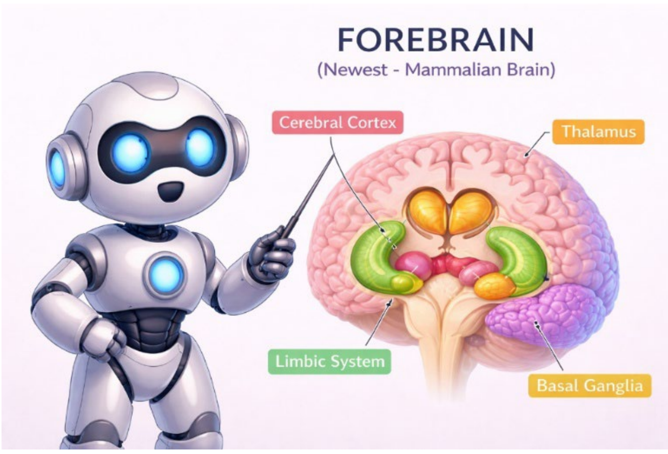


Figure 5.2: Threshold and Reset — Spike fires when membrane potential reaches threshold.

PART 2

COMPUTATIONAL NEUROSCIENCE

Modeling the Brain with Mathematics

🎯 WHAT YOU'LL LEARN IN PART 2

Now that you understand the brain's 'hardware,' we can learn how to model it mathematically.

In Part 2, you will learn:

- What computational neuroscience is and why it matters for AI
- The Leaky Integrate-and-Fire (LIF) model — a mathematical model of a neuron
- How to read and understand the LIF equation (don't worry — we'll go step by step!)
- How neurons encode information (rate coding, temporal coding, population coding)
- Why these biological insights matter for building better AI

Every equation will be explained in plain English. If you don't understand, go back and re-read!

📚 MATH ANXIETY? READ THIS.

Many students worry about the math. Here's how to approach it:

1. EVERY equation describes something physical that you learned in Part 1
2. We will explain EVERY symbol in plain English before using it
3. The goal is UNDERSTANDING, not memorization
4. If an equation confuses you, try drawing what it describes

Remember: the math is just a precise way of describing what you already learned. The neuron doesn't care about equations — it just does its thing. The equations are OUR tool for understanding and simulating that behavior.

SECTION 6

What is Computational Neuroscience?

6.1 Definition

Computational neuroscience is the field that uses mathematical models and computer simulations to understand how the brain processes information.

💡 **WHY SHOULD YOU CARE? Companies like Tesla, Intel, IBM, and BrainChip are hiring engineers who understand this! Neuromorphic chips (brain-inspired hardware) use 1000x less power than regular AI chips. The next iPhone might have a neuron-inspired processor. This isn't just academic — it's the future of AI!**

Instead of asking 'what chemicals does the brain use?' (biochemistry) or 'what structures does the brain have?' (anatomy), computational neuroscience asks:

What are the brain's ALGORITHMS?

This is exactly the question that matters for AI. If we can discover the algorithms the brain uses, we might be able to implement them in machines.

6.2 Why Use Math to Study the Brain?

Three key reasons:

1. THE BRAIN PROCESSES INFORMATION

Information processing can be described mathematically. When you recognize a face, your brain is performing computations — transforming pixel-like inputs into the category 'this is Mom.' Math is the language for describing such transformations precisely.

2. MODELS LET US TEST THEORIES

If someone says 'the brain works like X,' we can build a mathematical model of X, run simulations, and see if the results match what real brains do. If they don't match, the theory is probably wrong. Math turns vague ideas into testable predictions.

3. SIMULATIONS LET US EXPLORE THE IMPOSSIBLE

We can't record from all 86 billion neurons at once. We can't run experiments that would harm people. But we can simulate millions of model neurons and observe what emerges. Simulations let us ask 'what if?' questions we couldn't answer any other way.

6.3 David Marr's Three Levels (1982)

The cognitive scientist David Marr proposed that we can understand any information-processing system at three levels:

COMPUTATIONAL	WHAT problem is being solved? *Example: 'Recognize faces in images'*
ALGORITHMIC	HOW is it solved? What representations and steps? *Example: 'Detect edges → shapes → features → match to templates'*
IMPLEMENTATIONAL	What HARDWARE does it? What's the physical substrate? *Example: 'V1 → V2 → V4 → IT cortex (neurons, synapses, circuits)'*

Part 1 of this module gave you Level 3 (implementation) knowledge — you learned the brain's hardware. Part 2 focuses on Level 2 (algorithmic) — the mathematical descriptions of how that hardware processes information.

6.4 Why This Matters for AI Engineers

🤖 **FROM BRAINS TO MACHINES**

As AI engineers, computational neuroscience matters because:

BORROW ALGORITHMS: Deep learning already borrowed hierarchical feature detection from visual cortex. Reinforcement learning matches dopamine signaling. Experience replay came from hippocampus. What other algorithms are waiting to be discovered?

PROVEN SOLUTIONS: Evolution tested neural algorithms for hundreds of millions of years across countless species. When we find a computational principle that appears in fish, frogs, mice, and humans, it's probably a good solution to a fundamental problem.

SOLVE HARD PROBLEMS: The brain handles problems AI still struggles with:

- Energy efficiency (20 watts vs. megawatts)
- Learning from few examples (one-shot learning)
- Generalizing to new situations
- Common sense reasoning

Understanding HOW the brain solves these problems might help us build AI that can too.

SECTION 7

The Leaky Integrate-and-Fire Model

7.1 Why Model Neurons?

In Part 1, you learned that neurons receive inputs, integrate them, and fire spikes. Now we want to describe this behavior with mathematics so we can:

- Simulate neurons on computers
- Make precise predictions about neuron behavior
- Build brain-inspired AI systems (Spiking Neural Networks)

The Leaky Integrate-and-Fire (LIF) model is the simplest model that captures the essential behavior of real neurons. It's not perfect — real neurons are more complex — but it's a great starting point. (More detailed models like the Hodgkin-Huxley equations capture full ion channel dynamics but are computationally expensive — the LIF model gives us 90% of the insight with 10% of the complexity.)

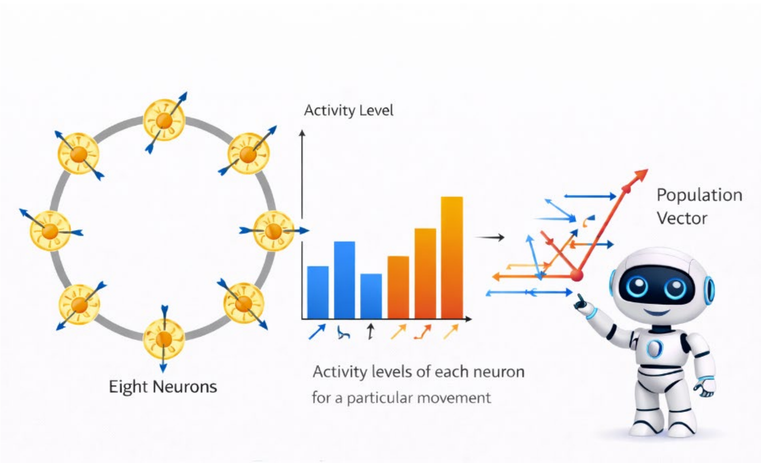


Figure 7.1: The LIF Model — Simplified neuron model capturing essential spiking behavior.

7.2 The Leaky Bucket Analogy

Before we see any equations, let's understand the LIF model through an analogy.

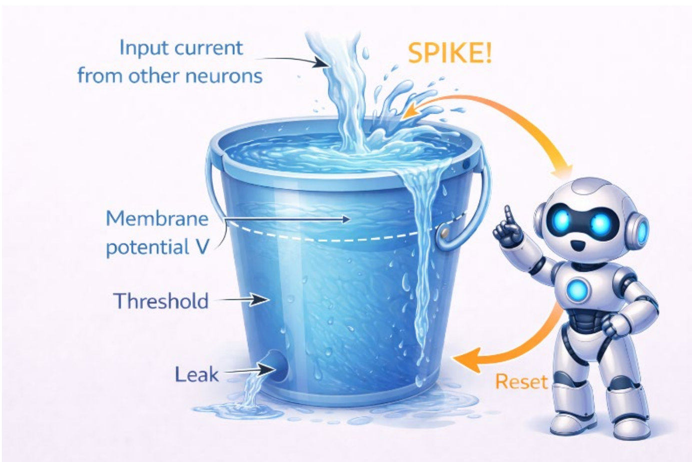


Figure 7.2: The Leaky Bucket Analogy — Intuitive model of LIF dynamics.

Think of a neuron as a bucket with a hole in the bottom:

WATER POURING IN = Input current from other neurons. More inputs = more water.

WATER LEVEL = Membrane potential (voltage). Higher water = voltage closer to threshold.

HOLE IN BUCKET = The 'leak.' Water constantly drains out, just like voltage constantly decays toward rest.

OVERFLOW LINE = Threshold. When water reaches this level, the bucket overflows.

OVERFLOW = SPIKE! The neuron fires. Then the bucket empties (reset) and starts filling again.

> 💡 **THE KEY INSIGHT**
>
> The neuron is a RACE between two forces:
>
> - LEAK: Always pulling voltage back toward rest (water draining out)
> - INPUT: Pushing voltage toward threshold (water pouring in)
>
> If input is weak → leak wins → water level stays low → no spike
>
> If input is strong → input wins → water overflows → SPIKE!
>
> The math simply describes this race precisely.

7.3 Understanding the Math (Step by Step)

😌 RELAX! We're about to look at some equations, but I promise they're easier than they look. Each piece has a simple meaning. Take your time. Read each step twice. By the end, you'll say 'Oh, that's it?' Let's go!

Now let's build up to the equation piece by piece. Don't skip ahead — each step builds on the last.

STEP 1: What Does dV/dt Mean?

You'll see 'dV/dt' in the equation. This is calculus notation, but you don't need calculus to understand it.

dV/dt simply means: 'How fast is V changing?'

🧪 *TRY THIS: Look at your phone battery right now. If it's at 67% and was at 70% an hour ago, then its 'dB/dt' (rate of battery change) is -3% per hour. See? You already understand rates of change — neurons work the same way, just with voltage instead of battery percentage!*

That's it. Think of everyday examples:

• Your phone battery level changes over time. The RATE tells you how fast it's charging or draining.

• Your car's position changes over time. The RATE of change is your speed.

• The membrane potential V changes over time. dV/dt tells you how fast.

If dV/dt is POSITIVE → V is increasing → voltage is rising toward threshold

If dV/dt is NEGATIVE → V is decreasing → voltage is falling toward rest

If dV/dt is ZERO → V is not changing → voltage is stable

STEP 2: What Does τ (tau) Mean?

The Greek letter τ (pronounced 'tau') is called the 'time constant.' It controls how SLOWLY things happen.

Think of τ like the size of the hole in our leaky bucket:

• LARGE τ (big number, like 20 ms) = Small hole = Water drains slowly = Neuron responds slowly, 'remembers' inputs longer

• SMALL τ (small number, like 5 ms) = Big hole = Water drains quickly = Neuron responds quickly, 'forgets' inputs faster

Typical values for real neurons: τ is about 10-20 milliseconds.

STEP 3: The Leak Term

The leak term is: -(V - V_rest)

What does this mean?

• V is the current voltage (membrane potential)

• V_rest is the resting voltage (typically -70 mV)

• (V - V_rest) is how far the voltage is from rest

• The negative sign means it OPPOSES deviation from rest

Example: If V = -60 mV and V_rest = -70 mV:

• (V - V_rest) = (-60) - (-70) = +10 mV (we're 10 mV above rest)
• -(V - V_rest) = -10 mV (the leak pushes us BACK DOWN toward rest)

The leak term always pushes voltage back toward resting. This is the 'leaky' in 'Leaky Integrate-and-Fire.'

STEP 4: The Input Term

The input term is: R × I

What does this mean?

- I is the input current (from other neurons or external stimulation)
- R is the membrane resistance (how much the membrane resists current flow)
- R × I is the voltage change caused by the input (Ohm's Law: V = IR)

More input current → bigger push toward threshold.

STEP 5: Putting It All Together

$$\tau \times dV/dt = -(V - V_{rest}) + R \times I$$

🎯 SUPER SIMPLE VERSION — Think of Your Bank Account:

• Your account balance = Voltage (V)
• $1000 baseline = Resting voltage (V_rest)
• Monthly bills drain money = Leak (pulls balance back toward baseline)
• Your paycheck adds money = Input current (pushes balance up)
• If balance hits $2000, you treat yourself! = Threshold (triggers a spike!)

The equation just says:
BALANCE CHANGE = (bills pulling down) + (paycheck pushing up)

That's it! The neuron equation works exactly the same way.

In plain English, this equation says:

How fast voltage changes = Leak pulling down + Input pushing up

Or even simpler:

Change in voltage = (Force toward rest) + (Force from input)

📝 READING THE EQUATION

Let's read τ × dV/dt = -(V - V_rest) + R × I piece by piece:

LEFT SIDE: τ × dV/dt

'The time constant τ times how fast V is changing'

Or: 'How the membrane potential changes, with some sluggishness'

RIGHT SIDE PART 1: -(V - V_rest)

'The negative of how far V is from rest'

This is the LEAK: always pushes V back toward rest

RIGHT SIDE PART 2: + R × I

'Resistance times input current'

This is the INPUT: pushes V toward threshold

THE WHOLE THING:

'V changes based on leak pulling it toward rest VERSUS input pushing it toward threshold'

STEP 6: The Threshold and Reset

The equation describes how voltage changes. But what happens when voltage reaches threshold?

THRESHOLD RULE: When V reaches V_threshold (typically -55 mV), the neuron FIRES A SPIKE.
RESET RULE: Immediately after firing, V is RESET back to V_rest (or slightly below).
REFRACTORY PERIOD: For a brief time after firing (1-2 ms), the neuron cannot fire again.

These rules are handled separately from the equation — they're 'if-then' conditions that the simulation checks at each time step.

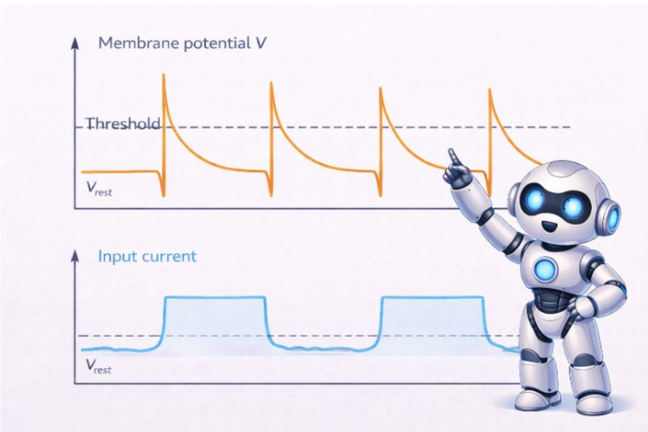


Figure 7.3: LIF Simulation Output — Membrane potential trace showing spikes.

7.4 A Worked Example

Let's trace through what happens with specific numbers. Don't just read this — work through it yourself!

📊 **EXAMPLE: Will the Neuron Fire?**

Given parameters:

- $\tau = 20$ ms (time constant)
- $V_{rest} = -70$ mV (resting potential)
- $V_{threshold} = -55$ mV (threshold)
- $R = 10$ MΩ (membrane resistance)
- $I = 2$ nA (input current)

Question: What is the steady-state voltage? Will the neuron fire?

At steady state, voltage isn't changing, so $dV/dt = 0$.
Plugging into $\tau \times dV/dt = -(V - V_{rest}) + R \times I$:
$0 = -(V - (-70)) + (10 \times 2)$

$0 = -(V + 70) + 20$

$0 = -V - 70 + 20$

$V = -50$ mV

Since -50 mV is ABOVE threshold (-55 mV), YES, the neuron will fire!

In reality, it won't reach -50 mV — it will fire as soon as it crosses -55 mV, then reset.

SECTION 8

Neural Coding — How the Brain Represents Information

8.1 The Coding Problem

Neurons communicate with spikes. But how do spikes represent information? When you see a red apple, how does your brain encode 'red' and 'apple' using just electrical pulses?

This is the neural coding problem. There are several solutions, and the brain probably uses all of them:

💡 **WHY SHOULD YOU CARE? Understanding neural coding is like understanding how to compress video files. Netflix uses smart encoding to stream movies with less data. Your brain does the same thing — it encodes your entire experience of reality using incredibly efficient spike codes. Learning this helps you build more efficient AI!**

8.2 Rate Coding

The simplest idea: information is encoded in the FIRING RATE — how many spikes per second.

Example: Touching your skin

• Light touch → neuron fires at 10 Hz (10 spikes per second)

EVERYDAY EXAMPLE: It's like how hard you press a video game controller. Light press = slow walking. Hard press = full sprint. Your thumb pressure (input strength) controls your character's speed (firing rate). Same principle!

• Medium pressure → neuron fires at 50 Hz

• Strong pressure → neuron fires at 100 Hz

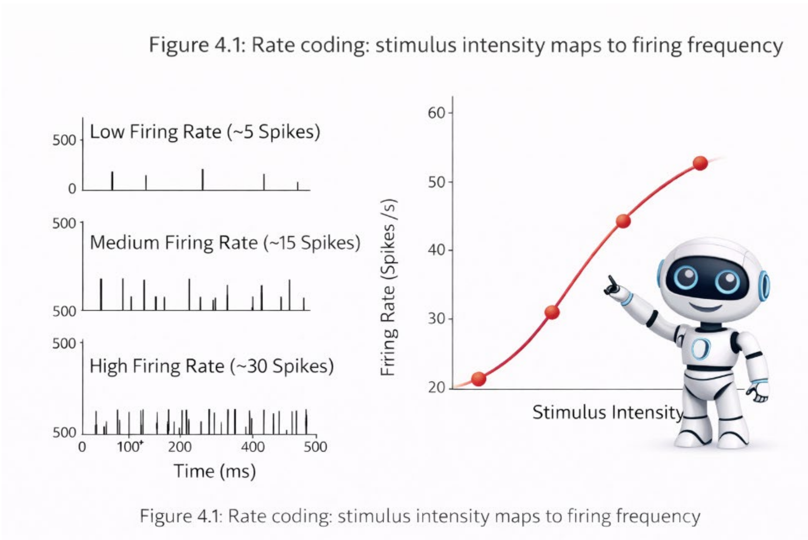


Figure 8.1: Rate Coding — Stimulus intensity encoded as firing rate.

Advantages:

• Simple and robust — even if some spikes are lost, the average rate is still informative

• Easy to decode — just count spikes

Disadvantages:

• SLOW — you need time to count enough spikes to estimate the rate (50-100 ms)

• Ignores timing information

> 🤖 **AI CONNECTION: ANNs Use Rate Coding**
>
> Standard artificial neural networks essentially use rate coding!
>
> When an ANN node outputs 0.73, you can interpret this as:
>
> 'This neuron would fire at 73% of its maximum rate.'
>
> The continuous values in ANNs approximate what rate coding captures.
>
> But real neurons also use timing information, which ANNs throw away...

8.3 Temporal Coding

A more sophisticated idea: information is encoded in the TIMING of spikes, not just the rate.

Types of temporal coding:

LATENCY CODING: Information is in how QUICKLY a neuron responds. Stronger stimuli → faster response → earlier first spike.

PHASE CODING: Information is in WHEN spikes occur relative to brain oscillations (like theta waves in hippocampus).

SYNCHRONY CODING: Neurons that fire TOGETHER might represent the same object. All the neurons responding to different features of an apple (red, round, shiny) synchronize their spikes to indicate 'these features belong together.'

Advantages:

• Potentially much faster than rate coding — a single spike's timing carries information

• Can encode more information in the same amount of time

Disadvantages:

• Requires precise timing — noise can corrupt the signal

• Harder to decode

🤖 **AI CONNECTION: Spiking Neural Networks**

Spiking Neural Networks (SNNs) exploit temporal coding!

Unlike standard ANNs, SNNs process discrete spikes over time.

They can use spike timing, not just rates, to encode and process information.

This makes SNNs potentially:

- More efficient (only compute when spikes occur)
- More powerful (temporal patterns add representational capacity)
- More biological (closer to how real brains work)

You'll build SNNs in Module 04!

8.4 Population Coding

Another key idea: information is encoded in the PATTERN of activity across many neurons, not in any single neuron.

💥 EVERYDAY EXAMPLE: Imagine asking 10 friends which restaurant to go to. No single vote decides — it's the PATTERN of all votes combined. '3 votes Thai, 5 votes Mexican, 2 votes pizza' → you go Mexican! Neurons 'vote' the same way.

💥 EVERYDAY EXAMPLE: Think of voting for where to eat dinner with 10 friends. No single person decides — you look at the PATTERN of all votes. '3 votes Thai, 5 votes Mexican, 2 votes pizza' → Mexican wins! Your brain's neurons 'vote' the same way to decide what you're seeing or doing.

Example: Encoding direction of movement

Each neuron has a 'preferred direction' it responds to most strongly. The activity follows a cosine tuning curve — maximum for the preferred direction, decreasing smoothly for other directions. But it also responds (less strongly) to similar directions. No single neuron precisely encodes direction, but the PATTERN across many neurons does.

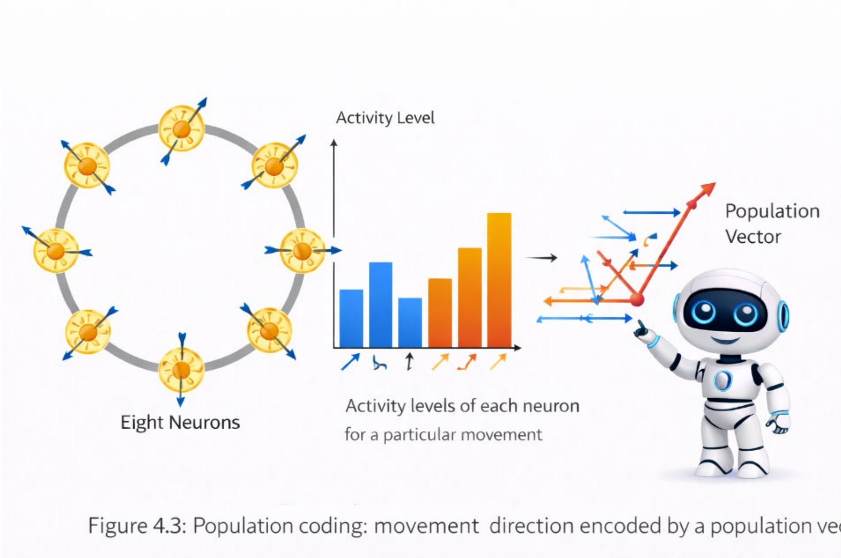


Figure 8.2: Population Coding — Multiple neurons encode direction through combined activity.

Advantages:

• Robust — no single neuron is critical (graceful degradation)

• Precise — many coarse-tuned neurons can achieve fine precision together

• Flexible — same neurons can encode many things depending on context

> 🤖 **AI CONNECTION: Embeddings Are Population Codes**
>
> In AI, embeddings (like word2vec or transformer embeddings) are population codes!
>
> A word isn't represented by a single number, but by a PATTERN across many dimensions (e.g., 768 dimensions in BERT).
>
> Each dimension is like a neuron with partial information.
>
> The pattern across all dimensions encodes meaning.
>
> This is why 'king - man + woman ≈ queen' works — the operations combine population codes in meaningful ways.

SECTION 9

Connecting Neuroscience to AI

9.1 Key Takeaways for AI Engineers

Let's summarize what we've learned and why it matters for AI:

1. Brain Organization Suggests Modular AI

The brain has specialized regions (vision in occipital, language in temporal, planning in frontal) but they all share similar cortical architecture. This suggests AI systems might benefit from modular designs with shared underlying algorithms.

2. Biological Neurons ≠ ANN Nodes

Real neurons have dynamics (the LIF equation), discrete spikes, and timing-dependent computation. Standard ANNs abstract this away. Spiking Neural Networks (Module 04) will show you how to preserve these features.

3. Multiple Coding Schemes = Multiple Representations

The brain uses rate, temporal, and population coding simultaneously. Current AI systems mostly use rate-like codes. Exploiting temporal coding is an open opportunity.

4. The Brain's Algorithms Work

Evolution tested neural algorithms for millions of years. When we find computational principles in brains (like TD learning in dopamine neurons, or hierarchical feature detection in visual cortex), they're worth taking seriously.

9.2 What's Next: Module 04

In Module 04, you will:

• Connect individual LIF neurons into networks

• Learn about synaptic plasticity and Spike-Timing-Dependent Plasticity (STDP) — the biological learning rule

• Train Spiking Neural Networks (SNNs) using surrogate gradients

• Explore neuromorphic hardware (Intel Loihi, IBM TrueNorth)

• Build an image classifier using snnTorch

The LIF neuron you learned about in this module is the building block. Module 04 shows you how to connect these building blocks into systems that can learn.

> 🌉 **BRIDGE TO MODULE 04**
>
> Module 03 gave you:
>
> - WHERE: Brain anatomy — regions and structures
> - HOW: Neuron computation — the LIF model
> - WHAT: Neural coding — rate, temporal, population
>
> Module 04 will add:
>
> - LEARNING: How neurons change their connections
> - NETWORKS: How neurons work together
> - HARDWARE: Neuromorphic chips that implement these principles
>
> Together: anatomy + computation + plasticity + networks = the foundation for neuromorphic AI

SECTION 10

Lab Preparation — Building Your First Neuron Model (Optional)

In the lab, you will implement an LIF neuron from scratch. Here's what you need to know:

10.1 The Simulation Loop

Your simulation will:

1. Start with V = V_rest

2. At each time step (dt, typically 0.1 ms):

  a. Calculate dV using the LIF equation

  b. Update V: V = V + dV × dt

c. Check if V ≥ V_threshold

d. If yes: record a spike, reset V = V_rest

3. Repeat for desired simulation time

💻 OPTION: USE WOLFRAM LANGUAGE (EASIER!)

Don't want to code from scratch? Use Wolfram! It handles the math automatically.

STEP 1: Go to wolframcloud.com You have a Wolfram account through HCC
STEP 2: Copy and paste this code:

```
(* LIF Neuron Simulation in Wolfram *)
tau = 20; (* time constant in ms *)
Vrest = -70; (* resting voltage *)
Vthresh = -55; (* threshold *)
R = 10; (* resistance *)
I = 2; (* input current *)

sol = NDSolve[{
 tau*V'[t] == -(V[t] - Vrest) + R*I,
 WhenEvent[V[t] > Vthresh, V[t] -> Vrest],
 V[0] == Vrest
}, V, {t, 0, 100}];

Plot[V[t] /. sol, {t, 0, 100},
 PlotLabel -> "LIF Neuron Membrane Potential",
 AxesLabel -> {"Time (ms)", "Voltage (mV)"}]
```

STEP 3: Click "Run" — instant neuron simulation!

🎮 INTERACTIVE VERSION: Add sliders to play with parameters:

```
Manipulate[
 Module[{sol},
  sol = NDSolve[{tau*V'[t] == -(V[t] - Vrest) + R*I,
    WhenEvent[V[t] > Vthresh, V[t] -> Vrest], V[0] == Vrest},
    V, {t, 0, 100}];
  Plot[V[t] /. sol, {t, 0, 100}, PlotRange -> {-80, -40}]
 ],
 {{tau, 20, "τ (time constant)"}, 5, 50},
 {{I, 2, "Input Current"}, 0, 5},
 {{R, 10, "Resistance"}, 1, 20}
]
```

This lets you SEE how changing τ, R, and I affects the neuron!

4. Plot the results

10.2 Exploration Questions

After your basic simulation works, explore:

1. What input current is required to make the neuron fire? (This is called the rheobase.)

2. How does firing rate change as you increase current above threshold? (This is the f-I curve.)

3. What happens when you change τ? Compare τ = 5 ms, 20 ms, and 50 ms.

4. How would you implement a refractory period? Add one and observe the effect on maximum firing rate.

5. What happens with noisy input current? Add random fluctuations and observe.

💻 WOLFRAM EXPLORATION OPTIONS

For each exploration question, here's how to do it in Wolfram:

1. FINDING RHEOBASE (minimum current to fire):

```
(* Try different I values until neuron fires *)
Table[{I, CountSpikes[I]}, {I, 0, 3, 0.1}]
```

2. PLOTTING THE f-I CURVE:

```
ListPlot[Table[{I, FiringRate[I]}, {I, 1, 5, 0.2}],
  AxesLabel -> {"Current", "Firing Rate (Hz)"}]
```

3. COMPARING DIFFERENT τ VALUES:

```
Plot[{V5[t], V20[t], V50[t]}, {t, 0, 50},
  PlotLegends -> {"τ=5ms", "τ=20ms", "τ=50ms"}]
```

4. ADDING NOISE:

```
I[t] = 2 + RandomReal[{-0.5, 0.5}]  (* noisy input *)
```

🔗 RESOURCES:

• Wolfram Demonstrations: demonstrations.wolfram.com (search "integrate and fire")
• Wolfram U Free Courses: wolfram.com/wolfram-u/
• Wolfram Community: community.wolfram.com (ask questions!)

📝 ASSIGNMENT OPTIONS:

You may complete the lab exercises using:

☐ Python (traditional approach)
☐ Wolfram Language (recommended for visualization)
☐ MATLAB (if you prefer)

All three will be accepted. Choose whatever helps you learn best!

SECTION 11

Glossary

Action Potential: A rapid electrical spike (~1-2 ms duration, ~100 mV amplitude) that propagates down the axon when membrane potential exceeds threshold. The fundamental unit of neural communication. All-or-nothing.

Axon: The long projection from a neuron's cell body that transmits output signals to other neurons. Can be myelinated for faster transmission.

Basal Ganglia: A group of subcortical nuclei involved in action selection, habit formation, and reward-based learning. Implements biological reinforcement learning via dopamine.

Cerebral Cortex: The brain's outer layer of neural tissue (~2-4 mm thick), organized into six layers and four lobes (frontal, parietal, temporal, occipital).

Computational Neuroscience: The field that uses mathematical models and computer simulations to understand how the brain processes information.

Dendrites: Branching extensions of a neuron that receive input from other neurons via synapses**.**

dV/dt: Mathematical notation meaning 'how fast V (voltage) is changing over time.' Rate of change of membrane potential. 🎯 SIMPLE: 'How fast is voltage changing?' — that's all it means!

Hippocampus: A subcortical structure crucial for forming new episodic memories and spatial navigation. Contains place cells.

Leaky Integrate-and-Fire (LIF) Model: A simplified neuron model: $\tau \times dV/dt = -(V - V_{rest}) + R \times I$. Captures essential spiking behavior.

Membrane Potential: The voltage difference between inside and outside of a neuron. Resting: ~-70 mV. Threshold: ~-55 mV.

Population Coding: Encoding information in the pattern of activity across many neurons, not single neurons.

Rate Coding: Encoding information in firing rate (spikes per second).

Refractory Period: Time after a spike during which a neuron cannot fire again (absolute) or requires stronger input (relative).

Synapse: The junction between two neurons where information is transmitted via neurotransmitters.

Temporal Coding: Encoding information in the precise timing of spikes, not just rate.

Thalamus: A subcortical structure that relays sensory information to cortex and gates information flow.

τ (tau): The membrane time constant. Controls how quickly a neuron responds to inputs. Larger τ = slower response. 🎯 SIMPLE: Think of it as 'how sluggish' the neuron is. Big τ = slow and steady. Small τ = quick and jumpy.

Threshold: The membrane potential (~-55 mV) at which a neuron fires an action potential.

SECTION 12

References

Textbooks

Bear, M., Connors, B., & Paradiso, M. (2020). Neuroscience: Exploring the Brain (5th ed.). Jones & Bartlett Learning.

→ *Comprehensive neuroscience textbook; Chapters 2-4 cover neuron biology.*

Carter, R. (2019). The Human Brain Book: An Illustrated Guide to its Structure, Function, and Disorders. DK Publishing.

→ *Excellent visual reference for brain anatomy.*

Gerstner, W., Kistler, W., Naud, R., & Paninski, L. (2014). Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition. Cambridge University Press.

→ *Free online at neuronaldynamics.epfl.ch; excellent coverage of the LIF model.*

Hawkins, J. (2021). A Thousand Brains: A New Theory of Intelligence. Basic Books.

→ *Accessible introduction to the Thousand Brains Theory.*

Key Papers

Hassabis, D., et al. (2017). Neuroscience-inspired artificial intelligence. Neuron, 95(2), 245-258.

Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields in the cat's visual cortex. Journal of Physiology.

Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. Science.

Online Resources

Neuroscience Online (neuroscience.uth.edu) — Free open-access textbook

Neuronal Dynamics (neuronaldynamics.epfl.ch) — Free online textbook with Python exercises

3Blue1Brown Neural Networks (youtube.com/3blue1brown) — Excellent visualizations

Wolfram Language & Neural Networks (wolfram.com/language/neural-networks) — Interactive neuroscience simulations

Wolfram Demonstrations Project (demonstrations.wolfram.com) — Search 'neuron' for ready-to-use simulations

Image Credits

All illustrations in this booklet were created using ChatGPT (DALL-E) specifically for this course.

📋 SUPER QUICK SUMMARY — What You Learned:

✓ BRAIN = Three layers (hindbrain/midbrain/forebrain) + four lobes + subcortical structures
✓ NEURON = Dendrites (input) → Soma (processor) → Axon (output) → Synapse (connection)
✓ SPIKE = All-or-nothing electrical pulse when voltage hits threshold
✓ LIF MODEL = $\tau \times dV/dt = -(V - V_{rest}) + R \times I$ (Leak + Input = Change)
✓ CODING = Rate (how many spikes), Temporal (when spikes), Population (pattern across neurons)

That's it! You now understand how brains compute. 🧠✨

— End of Module 03 —

Next: Module 04 — Spiking Neural Networks